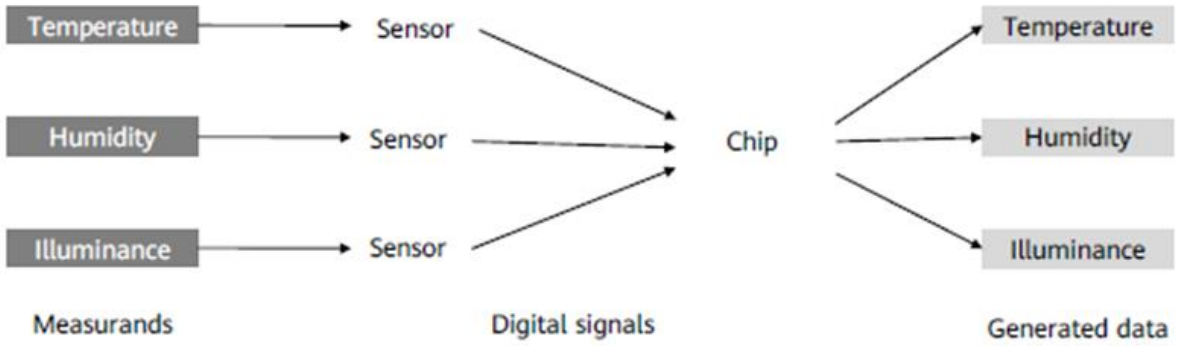
CHAPTER 4
DATA COLLECTION TECHNOLOGIES

Sensing Techonologies

Information Techonology is built on tree main pillars: computers, communications, and sensing technologies. Sensing technologies are an important indicator to measure a country's informatization level. Sensing technology refers to sensors that can sense something about the ambient environment or that detect some special substance.

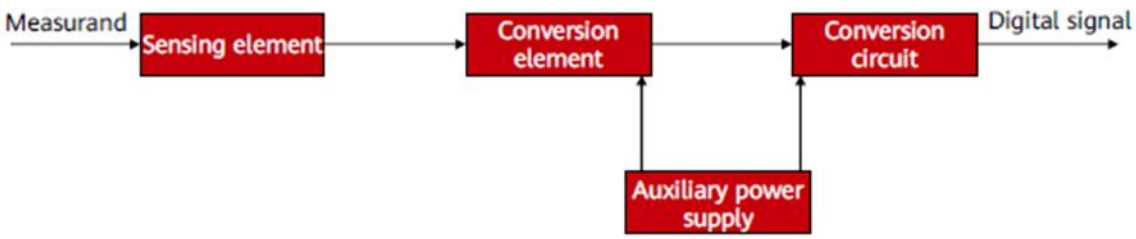


- Sensors

A sensor is a device that senses a measurand and converts it into a usable signal based on certain rules.

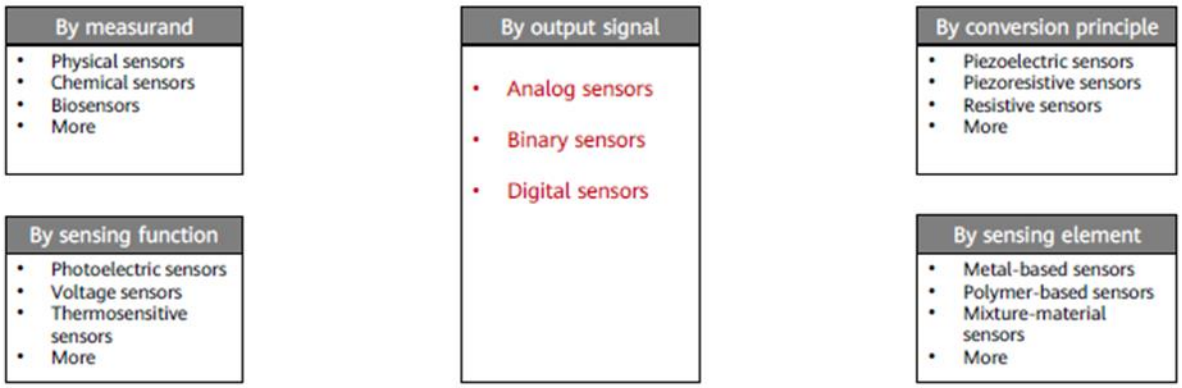
- Application Scenarios

Sensors are used in the military, energy, robotics, automatic control, environmental protection, transportation, healthcare, household appliances, remote sensing, and chemicals industries. IoT devices collect a lot of data. Device data comes from sensors.

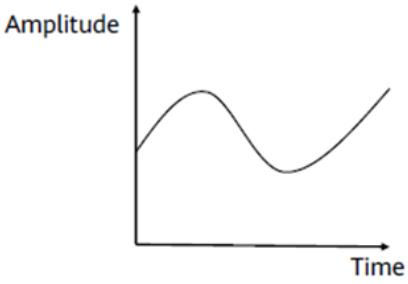


- Sensor Classification

Sensors can be classified using different methods:



Analog Sensors



Analog sensors output continuous signals and present measurand values using voltage, current, and temperature. The figure shows the current signal output by a thermistor. Any value on the curve represents a temperature value.

Binary Sensors

Signals output by binary sensors are not continuous. Binary sensors reflect two status signals: open and closed, similar to 1 and 0 of digital sensors. They are the most typical type of sensors and are used to detect object statuses, actions, or positions.

Digital Sensors

Digital sensors transform the A/D conversion modules of analog sensors to output digital signals. A digital sensor contains an amplifier, analog-to-digital converter (ADC), microprocessor, memory, communication interfaces, and temperature detection circuit. Digital sensors convert analog quantities into digital quantities in four steps: sampling, holding, quantization, and encoding. Analog-to-digital conversion formula:

$$A/D = \frac{U_A}{V_{DD}} \cdot 2^n = \frac{2^n}{V_{DD}} \cdot U_{A|}$$

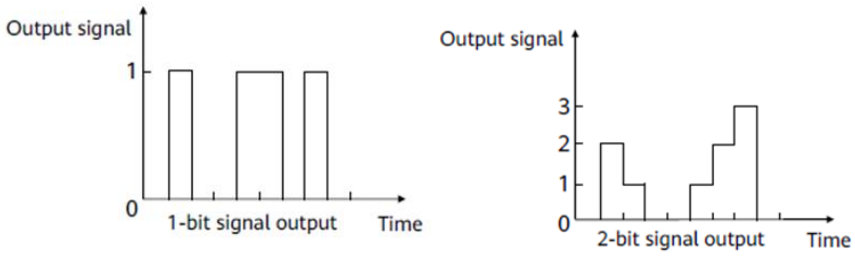
Where

A/D: digital quantity

UA: analog quantity

n: ADC resolution (number of bits)

VDD: power supply voltage of the conversion circuit



Working Principles of Common Sensors

- Temperature Sensors

Temperature sensors sense temperatures and convert them into usable signals. Based on the characteristics of the sensing materials and electronic components used, temperature sensors can be classified as resistance temperature detectors (RTDs) or thermocouple sensors.

- Humidity Sensors

Humidity sensors can sense environmental humidity changes and convert non-electrical quantities into electrical ones by detecting physical or chemical changes. They can be classified as resistive or capacitive sensors.

- Photoelectric Sensors

Photosensitive sensors use photosensitive elements to convert optical signals into electrical signals. Their physical basis is the photoelectric effect. Many electrical features of semiconductor materials change when exposed to light. There are three types of photoelectric effects: external, internal, and photovoltaic.

- Infrared Sensors

An infrared sensor consists of an optical system, a detecting element, and a conversion circuit. Optical systems can be classified based on their structure as either transmission or reflection system. Detecting elements can be classified as thermosensitive or photoelectric elements, depending on how they work.

Tag Identification Technologies

Barcode Identification

Barcode Identification Technologies

- IoT implements global goods tracking and information sharing, greatly improving management and operations efficiency. It affects each process of global supply chains, involving manufacturing, warehousing, logistics, goods tracking, retail, and public service industries. Automatic identification technologies using barcodes and electronic tags are a cornerstone of IoT. They enable objects to speak.
- One-dimensional (1D) barcodes are based on laser identification and two-dimensional (2D) barcodes are based on image identification. They are two types of barcode information collection technologies. Radio frequency identification (RFID) is a radio information collection technology

1D Barcodes

- Barcodes are a type of automatic identification technology. They remove bottlenecks in data recording and collection and are widely used for materials and production management.
- A barcode is an image consisting of black lines (bars) and white lines (spaces) with different widths arranged based on special encoding rules. It contains information based on these highly contrasting parallel lines.
- Barcode scanners identify the light reflected by barcodes based on the photoelectric effect. The spaces reflect strong signals, and the bars reflect weak signals.

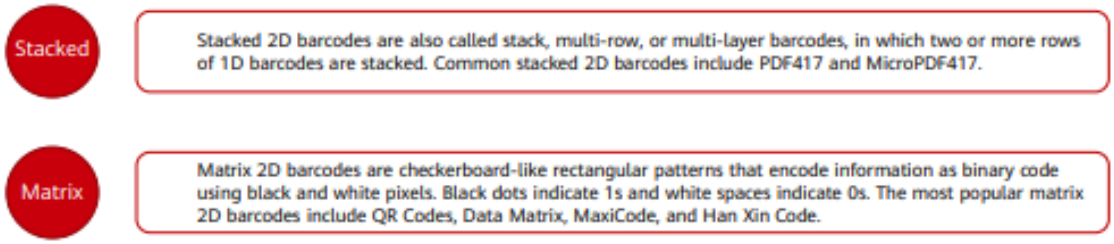
Barcode Type	Description
EAN-13	Used for products globally. The code contains 13 digits and supports numbers 0–9. Notches are included.
EAN-8	Used for products globally. The code contains 8 digits and supports numbers 0–9. Notches are included.
UPC-A	Used for products in the US and Canada. The code contains 12 digits and supports numbers 0–9. Notches are included.
EAN-128	All ASCII characters are supported. Its length is theoretically unlimited. It is the standard for the EAN/UCC system and is used for logistics units.
Code 39	Supports uppercase letters A–Z, numbers 0–9, and 43 characters. It can encode variable-length data and is mainly used for logistics tracking and production line workflow.
Code 128	128 ASCII characters are supported. Its length is theoretically unlimited. Code 128 has code sets A, B, and C.



Code-128 example

2D Barcodes

- 2D barcodes are graphical images that store information horizontally and vertically using black and white cells. The information density of a 2D barcode is several times that of a 1D barcode, so they can hold much more information.
- 2D barcodes can be classified by their encoding method as either stacked or matrix barcodes.



- 1D barcodes contain less information. As technology advanced, we developed 2D barcodes. They can represent more information in a smaller space.
- 2D barcodes not only identify a group of items but also describe a specific item. This is a great step forwards for barcode technologies. 2D barcodes also provide alignment patterns and error correction. Alignment patterns allow 2D barcodes to be scanned in any direction. Error correction enables correct identification and restoration of partial or damaged barcodes.
- 2D barcodes feature large storage capacity, high security, high traceability, strong resistance to damage, high availability, and cost-effectiveness. They can be used in forms, for information security, for tracking codes, permits and other certificates, stocktaking, and document backup.

Barcode Comparison

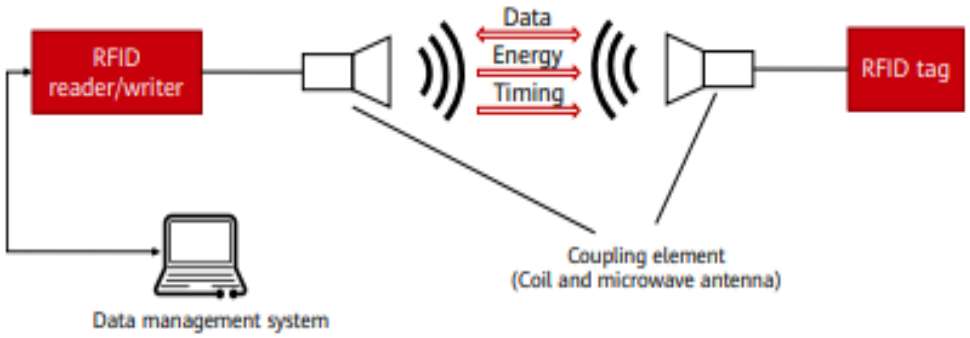
Item	1D Barcode	2D Barcode
Space utilization	Low (horizontal only)	High (horizontal & vertical)
Capacity	About 30 characters	1,850 characters or more
Information content	Letters and numbers	Letters, numbers, symbols, and figures
Expression method	1D barcodes cannot present product information directly. They need to connect to a database.	2D barcodes can present product information directly. They do not need to connect to a database.
Error correction	Not supported. Damaged barcodes cannot be read.	Supported. 2D barcodes can be read from any direction. At the highest error correction level, complete information can still be read from a 2D barcode that is only 50% complete.
Code systems	Code 128, EAN, ISBN, etc.	QR codes, PDF417, etc.
Application scenarios	Workflow management, transportation, warehousing, postal services, etc.	Mobile payments, website links, account login, information reading, e-commerce, and anti-counterfeiting, etc.

1D barcodes connect to databases where product data is managed. They are necessary for large retailers because they can improve stocktaking accuracy and efficiency.

RFID

RFID Overview

- RFID is a contactless automatic identification technology that uses radio waves for two-way data communications. Recording media (electronic tags or radio frequency cards) can be read or written to identify targets and exchange data.
- An RFID system consists of a reader, an RFID tag, and a data management system.

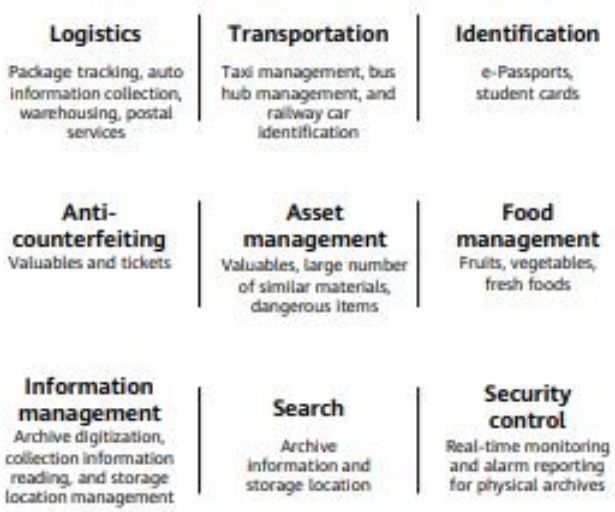


- A reader reads information from or writes information into tags. It can be a read/write device depending on the structure and technology used. It is an information control and processing center of the RFID system. When the RFID system is working, the reader emits radio frequency energy within an area to form an electromagnetic field, and the size of the area depends on the transmit power. When the tag in the area covered by the reader is triggered, data stored in the tag is sent or modified based on reader instructions. A reader can communicate with computer networks using APIs. A reader usually consists of a transceiver antenna, frequency generator, phase-locked loop, modulation circuit, microprocessor, memory, demodulation circuit, and peripheral interface.
- An electronic tag consists of a transceiver antenna, an AC or DC circuit, a demodulation circuit, a logic control circuit, a memory, and a modulation circuit

RFID Application Scenarios

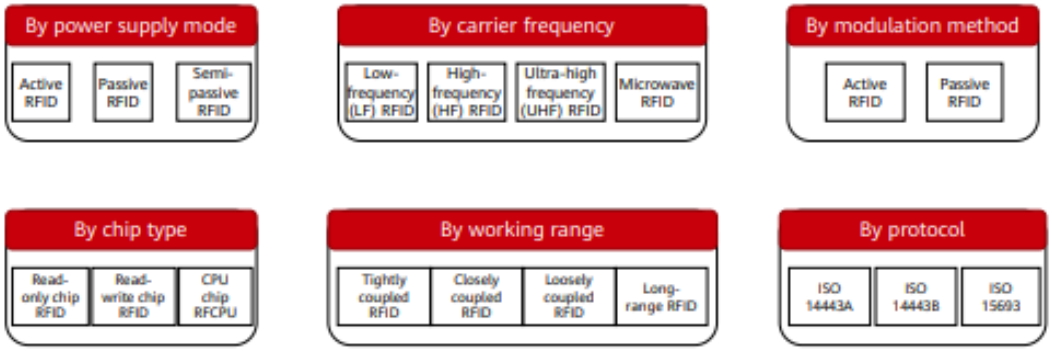






RFID Classification

- RFID tags can be classified into the following categories based on their power supply, carrier frequency, modulation method, working range, and chip type :

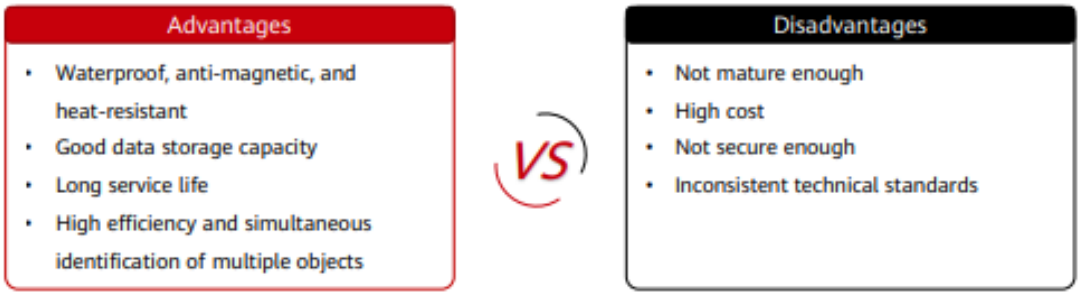


- Passive RFID

  Passive RFID is the earliest, most mature, and most widely used RFID system. A passive RFID tag receives microwave signals from an RFID reader and obtains energy through an electromagnetic induction coil to supply power to itself for a short time, so as to complete information exchange. As a passive RFID tag does not have the power supply system, its size can be reduced to cubic centimeters. It is cost effective and has a simple structure, low failure rate, and long service life. However, the effective identification range of passive RFID tags is short, so they are usually used for short-range identification. Passive RFID works at low frequency bands, such as 125 kHz and 13.56 MHz. Its typical applications include bus cards, ID cards, and dining cards.
- Active RFID

  Active RFID is an emerging technology, but it has been widely used and become a must in various fields, for example, in the electronic toll collection system for expressways. An active RFID has a power source and sends signals to an RFID reader actively. It has a large size, long transmission range, and high transmission speed. A typical active RFID tag can connect to an RFID reader 100 meters away, with a read rate of up to 1,700 reads per second. Active RFID works at high frequency bands, such as 900 MHz, 2.45 GHz, and 5.8 GHz. It can identify multiple tags at a time. As active RFID supports long-range transmission and is highly efficient, it is indispensable in scenarios that require high performance and long-range transmission

RFID Advantages and Disadvantages



- Advantages:
    1. RFID can update existing information more conveniently while reducing workforce, material resources, and financial resources.
    2. RFID uses computers to store information. Its maximum capacity is several megabytes, so it can store a large amount of information. This ensures smooth work.
    3. RFID has a long service life. It can be reused with careful handling.
    4. RFID makes information processing more convenient and enables simultaneous identification of multiple objects, greatly improving efficiency.
- Disadvantages:
    1. It is not mature enough. RFID is an emerging technology, which is not very mature. As UHF waves can bounce off metal objects and liquids, it is difficult to use UHF RFID tags for metal and liquid products.
    2. Its cost is high. RFID tags are dozens of times more expensive than common barcodes. Using many RFID tags will cause a high cost. This has made them less popular to some degree.
    3. It is not secure enough. Major security issues faced by RFID are unauthorized reading and tampering of RFID tag information
    4. Its technical standards are inconsistent.

Comparison of Automatic Identification Technologies

Item	Barcode	RFID
Reading methods	A scanner is required for reading the data.	Data can be read without a light source or even through packaging.
Reading rate	Only one barcode can be read at a time.	Multiple tags can be identified at once.
Data capacity	Small. A few dozen to a few thousand characters are allowed.	Large. Dozens of kilobytes are allowed.
Effective identification range (onsite)	Short	Long (active RFID)
Service life	Short. Information cannot be collected from incomplete or damaged barcodes.	Long. It can be used in polluted and radioactive environments with dusts and oil stains based on wireless communications.
Flexibility	Low. Tag data cannot be modified after being printed.	High. Tag data can be dynamically changed.
Communications mode	Supports only one-way communications. It cannot provide instant information or Internet data transmission.	Supports real-time communications. As long as an object with an RFID tag is within the reader detection range, its location can be dynamically tracked.
Cost	Low	High (if used in large quantities)

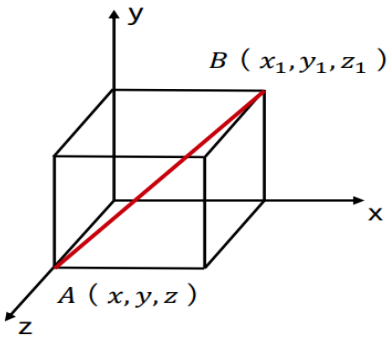
- Barcodes and RFID have similar application scenarios. When a company chooses from the two technologies, costs are a key factor. The cost of barcodes includes the costs of paper and ink. An active RFID tag with a memory chip costs more than USD2, and a passive RFID tag costs more than USD1. From the technical perspective, RFID tags have more advantages than barcodes.

Location Data Collection Technologies

Satellite Positioning

- Satellite positioning uses artificial earth satellites for precise positioning. It provides military and civilian users in air, sea, ground, or space with high-precision position, velocity, and time data in all weather conditions.
- There are four main global navigation satellite systems (GNSS): BeiDou Navigation Satellite System (BDS), Global Positioning System (GPS), Galileo Satellite navigation System (Galileo), Global Navigation Satellite System (GLONASS).
- Satellite positioning is commonly used for vehicle navigation, atmospheric physics observation, sailing route planning, missile guidance, and more.

Working Principles of Satellite Positioning



- Before calculation, you need to obtain each satellite's accurate time t0 and locations x0, y0, and z0. With these four elements, you can calculate the distance between a satellite and the target object using the 3D Pythagorean theorem formula
- In the figure on the right, the distance s between A and B is calculated as follows:

$$s = \sqrt{(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2}$$

- Electromagnetic waves travel at the speed of light, c. If data transmitted by a satellite at t0 is received by the target object at t, the transmission time is t–t0. Therefore, the preceding formula can be written as follows:

$$s = c * (t - t_0) = \sqrt{(x - x_1)^2 + (y - y_1)^2 + (z - z_1)^2}$$

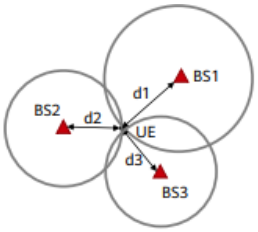
- There are three unknown numbers: x, y, and z. Theoretically, as long as signals of three satellites are received, you can calculate the remaining values using this equation. Devices often use crystal oscillators, which have limited precision, as clocks, so there is a concept called clock difference. In this case, another unknown number is introduced. This requires four satellites to be locked at the same time to locate the target object.

Base Station Positioning

- Base station positioning uses mobile telecommunication base stations to locate end users of carrier networks. The positioning service based on base stations is called location-based service (LBS).

- Although satellite positioning can provide precise positioning for positioning apps to locate terminals, only base station positioning works where satellite signals are unavailable, for example, in tunnels, underground parking lots, and roads under viaducts. Base station positioning and satellite positioning need to be used together to provide services in different outdoor scenarios.

Technical Principles of Base Station Positioning



- Base station positioning methods can be network-based or terminal-based. Each method includes many different positioning principles. Of these principles, trilateration is most commonly used.
- Trilateration:
    - The signal becomes weaker when it is farther away from a base station, so the signal strength received by a mobile phone can be used to estimate the distance between the mobile phone and the base station.
    - A base station is unique within a mobile network and has a unique geographical location. If you have three base stations, the distances between the mobile phone and each base station can be obtained. Then you just draw three circles with a base station as the center and the corresponding distance as the radius, and the phone will be at the point where the three circles intersect.

Network-based positioning

▫ Technology used in 2G, 3G, and 4G networks: time of arrival (TOA) positioning

▫ Technology used in 5G networks: angle of arrival (AOA) positioning under massive multiple input multiple output (MIMO)

• Terminal-based positioning

▫ Base ID positioning

▫ Trilateration positioning

▫ Positioning based on scenario analysis

Comparison of Outdoor Positioning Technologies

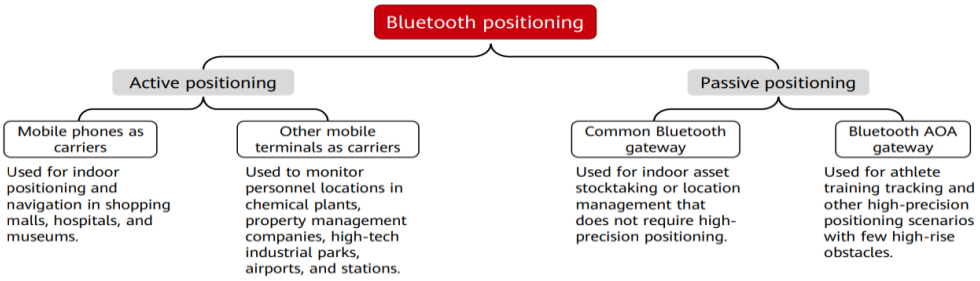
Item	Satellite Positioning	Base Station Positioning
Precision	High precision (5–10 m)	Low precision (20–200 m in urban areas and 1,000–2,000 m in suburban areas)
Power consumption	High. Terminals need to provide high-voltage power for GPS modules.	Low. Only base stations need to collect data.
Advantages	High-precision outdoor positioning and wide coverage	Fast positioning; Low power consumption; Unaffected by weather or high-rise buildings and not location dependent.
Disadvantages	High cost; High power consumption; The antenna of the GPS system must be placed outdoors and under the open sky. Otherwise, positioning will fail. It is affected by weather and location dependent.	Low precision The terminal has a registered SIM card and can receive signals from at least three base stations.

Wi-Fi Positioning

- Wi-Fi positioning is the most popular positioning method. It can be used for large-scale positioning, detection, and tracking in places with Wi-Fi networks. Network node selfpositioning is the foundation of Wi-Fi positioning.
- Wi-Fi positioning uses the k-nearest neighbor (KNN) algorithm. The hotspot or base station that the terminal is closest to is used as the location of the terminal. If there are multiple access points (APs), the received signal strength indicators (RSSIs) of three APs can be used for trilateration positioning

Bluetooth Beacon Positioning

- As Bluetooth is a very commonplace technology, many projects use Bluetooth beacons for positioning.
- Bluetooth beacon positioning uses a principle and approach similar to Wi-Fi positioning but delivers higher precision



Ultra-Wideband Positioning

- Ultra-wideband (UWB) positioning uses an anchor node (master device) and bridge nodes (at least two slave devices) and a blind node (tag to be located). The anchor and bridge nodes are deployed at known locations. Then, triangulation or fingerprinting method is used for positioning.
- UWB signals can easily penetrate common obstacles to obtain individual or object location data. They can also be used for precise indoor positioning and navigation in many scenarios, including tunnels, prisons, chemical plants, factories, coal mines, construction sites, power plants, nursing homes, exhibition halls, vehicles, equipment rooms, and airports.

Comparison of Indoor Positioning Technologies

Item	Wi-Fi Positioning	Bluetooth Beacon Positioning	Ultra-Wideband Positioning
Frequency range	2.4 GHz and 5 GHz	2.4 GHz	3.1–10.6 GHz
Transmission rate	Up to 1 Gbit/s	Up to 2 Mbit/s	Up to 500 Mbit/s
Latency	3–5s	3–5s	Less than 1 ms
Transmission range	Preferably within 50 m, but up to 100 m	Preferably within 25 m, but up to 100 m	Preferably within 50 m, but up to 200 m
Positioning precision	Less than 10 m	Bluetooth RSSI: 5–10 m Bluetooth AOA: 0.5–1 m	Less than 50 cm
Security	High	High	Relatively high
Penetration	Strong	Weak	Strong
Anti-interference	Strong	Weak	Strong
Power consumption	High	Relatively low	Medium
Cost	High	Low	High